

Write a program to implement Naïve Bayes algorithm in python and to display the results using confusion matrix and accuracy.

Code

```
import pandas as pd

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score, confusion_matrix, classification_report


# -----
# Load Iris Dataset
# -----

iris = load_iris()

X = iris.data
y = iris.target

# Create DataFrame (Optional)
df = pd.DataFrame(X, columns=iris.feature_names)
df['Target'] = y

print("First 5 Records:\n")
print(df.head())

# -----
# Split Dataset
```

```
# -----  
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.3, random_state=42  
)  
  
# -----  
# Train Naïve Bayes Model  
# -----  
model = GaussianNB()  
model.fit(X_train, y_train)  
  
# -----  
# Prediction  
# -----  
y_pred = model.predict(X_test)  
  
# -----  
# Results  
# -----  
print("\nActual Labels:")  
print(y_test)  
  
print("\nPredicted Labels:")  
print(y_pred)  
  
print("\nAccuracy:", accuracy_score(y_test, y_pred))  
  
print("\nConfusion Matrix:")
```

```
print(confusion_matrix(y_test, y_pred))
```

```
print("\nClassification Report:")
```

```
print(classification_report(y_test, y_pred, target_names=iris.target_names))
```

```
# -----
```

```
# Predict New Sample
```

```
# -----
```

```
new_sample = [[5.1, 3.5, 1.4, 0.2]]
```

```
prediction = model.predict(new_sample)
```

```
print("\nNew Sample Prediction:")
```

```
print("Predicted Class:", iris.target_names[prediction[0]])
```

First 5 Records:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	\
0	5.1	3.5	1.4	0.2	
1	4.9	3.0	1.4	0.2	
2	4.7	3.2	1.3	0.2	
3	4.6	3.1	1.5	0.2	
4	5.0	3.6	1.4	0.2	

Target

0	0
1	0
2	0
3	0
4	0

Actual Labels:

```
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]
```

Predicted Labels:

```
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 2 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]
```

Accuracy: 0.9777777777777777

Confusion Matrix:

```
[[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]
```

Classification Report:

	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	19
versicolor	1.00	0.92	0.96	13
virginica	0.93	1.00	0.96	13
accuracy			0.98	45
macro avg	0.98	0.97	0.97	45
weighted avg	0.98	0.98	0.98	45

New Sample Prediction:

Predicted Class: setosa

